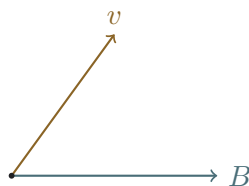


What Kind of Force Depends on Motion?

Name: _____

A charge beside a magnet, a question that sounds almost like a category error, and a law that only makes sense once several stranger ones have been asked first.



The question. Every force met so far has depended on where things are. How can a force depend on how fast something is moving?

Working rule: No formula is supplied until stage six. No assumption is made that force looks the same to every observer.

1–5. Five observations, in order

a Socratic ladder

Predict the outcome of each stage before working through it.

Stage	Setup	Prediction
1	Charge stationary beside a magnet	
2	Charge set moving, any direction	
3	Same direction, greater speed	
4	Velocity parallel to the field	
5	Velocity perpendicular to the field	

6. Only afterwards, reveal the law

the cross product

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B} \qquad |F| = qvB \sin \theta$$

Check the formula against each of your five predictions above. Does it agree with all five?

Why does the force appear here as a symbol for “into” or “out of” the page rather than as an arrow drawn in the plane of v and B ?

7. Whose force is it?

observer-dependence

An observer moves alongside the charge, matching its velocity exactly.

What that observer sees

Does that observer measure a magnetic force?

Two observers can disagree about whether a force acts on the same charge at the same instant. What does that disagreement tell you about electric and magnetic fields?

8. Core results

what this enquiry actually earned

The relation established, and only after the evidence.

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B} \qquad |F| = qvB \sin \theta$$

Concepts earned, in order.

- A force depending on velocity, not just position, was strange enough to earn five separate stages before any formula appeared.
- Zero force when parallel, maximum when perpendicular, is exactly what a sine dependence on angle predicts.
- The force is always perpendicular to the velocity, which is why it reads as a rotation out of the plane rather than an arrow within it.
- Force is not guaranteed to be the same for every observer; a charge at rest to one observer and moving to another can genuinely disagree about whether a force acts.
- Electric and magnetic fields are not two separate ingredients of nature but one field, sliced differently by observers moving at different velocities.

Every substantial claim should be accompanied by a reason, a right-hand-rule check, or a counterexample.